DATA ANALYSIS · FRESH FROM THE GRINDER

Coffee Sales Report

Data quality, exploratory analysis and business insights from every cup rung up between 01 March 2024 and 23 March 2025.

TRANSACTIONS	TOTAL SALES VALUE	AVERAGE TICKET	MEDIAN TICKET	DUPLICATES REMOVED
3,547	112,245.58	31.65	32.82	0

THE OVERVIEW

Today's brew

The supplied file contains **3,547 transaction records** across **8 coffee products**. The recorded sales value totals **112,245.58** in the source's unspecified monetary units, with an average transaction value of **31.65**.

The data spans **01 March 2024 to 23 March 2025**.

BEHIND THE COUNTER

Data quality & cleaning

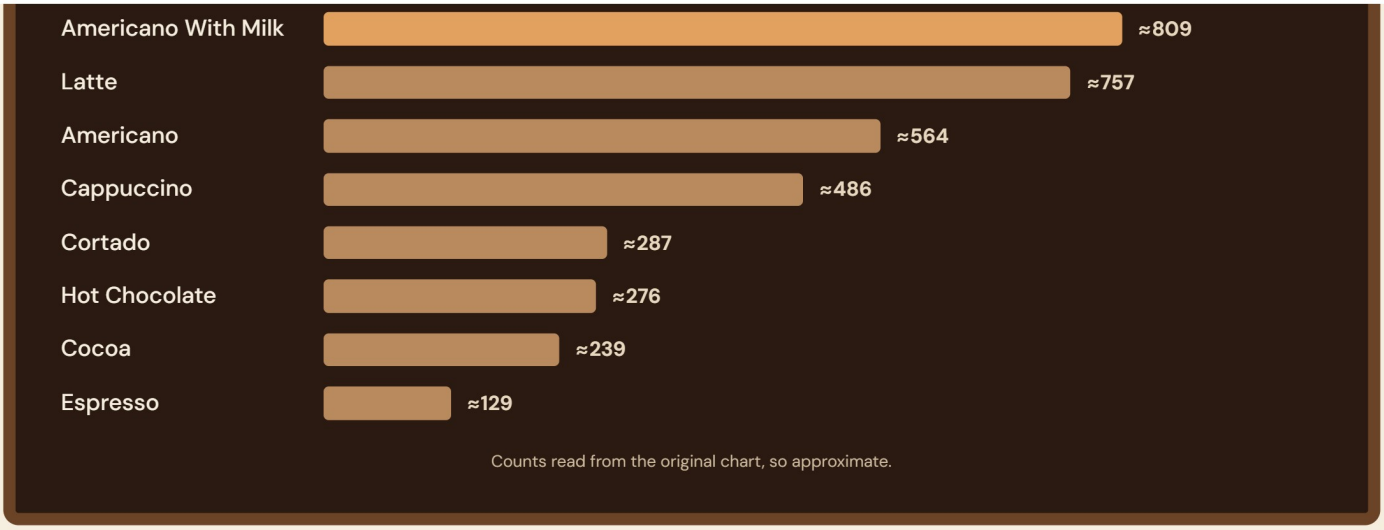
Initial inspection found **0 missing cells** and **0 exact duplicate rows**. Nothing needed throwing out.

- ✓ Column names standardized to lowercase snake_case; text trimmed and category labels normalized.
- ✓ Hour and sort fields and sales amount converted to numeric types; the date field parsed as a datetime.
- ✓ A transaction timestamp derived from date and time.
- ✓ The IQR review flags **0 transaction amounts** outside the fences (16.16 to 47.52). Unusual values can be valid and the source gives no price rules to verify, so none were removed.

PRODUCT PERFORMANCE

The menu board

Transactions by coffee product

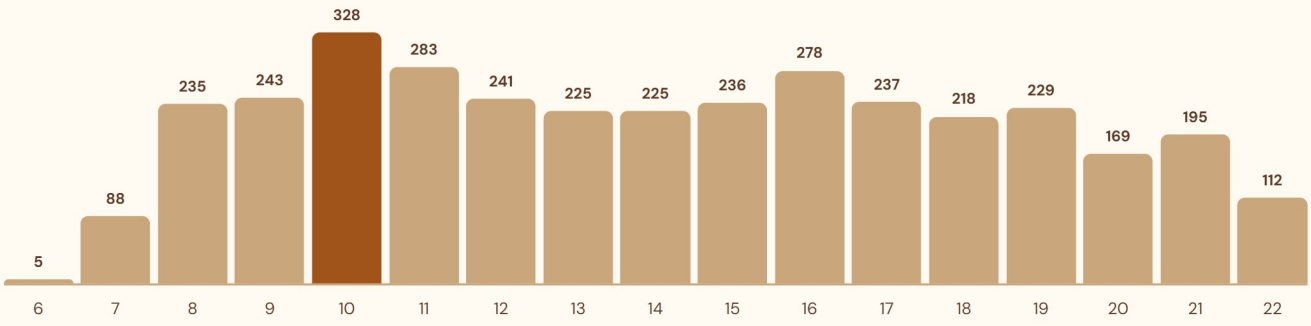


FINDING **Americano With Milk** is the most frequently purchased product in this dataset. The chart ranks products by transaction count; it should not be read as a profit ranking because cost and margin data are absent.

WHEN DO TRANSACTIONS OCCUR?

Rush hour

Transaction activity by hour



Hour of day · counts read from the original chart, so approximate

FINDING The busiest recorded hour is **10:00**. This pattern can help inform staffing and preparation schedules, subject to seasonality and operating-hour context.

PAYMENT MIX

Card, every time

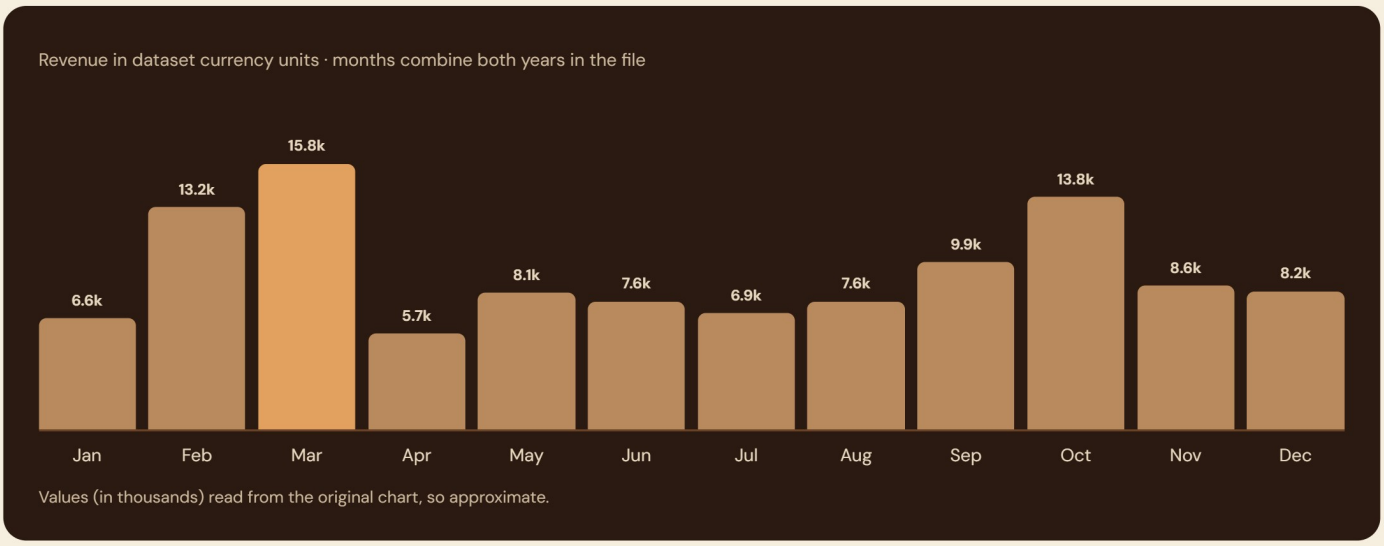
Card is the most frequently used payment type, covering all 3,547 transactions. This is a descriptive share, not evidence of payment preference among unique customers.

Payment receipt

Card	3,547
Other types	0
Card share	**100.0%**

MONTHLY SALES VALUE

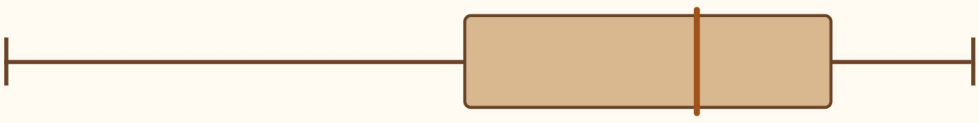
Revenue by month



Monthly comparisons may be affected by incomplete months or the dataset's limited coverage.

TRANSACTION VALUE DISTRIBUTION

What a typical ticket looks like



Middle half of tickets: **27.92 to 35.76** | Median: **32.82** | Whiskers: about 18.2 to 38.7 (read from chart)

FINDING The median transaction value is **32.82**, and the IQR method flags **0 potential outliers**. Records that do get flagged should be investigated rather than deleted automatically.

NEXT STEPS

Key recommendations

1. Use the busiest hours to review staff coverage and product preparation capacity.
2. Monitor high-volume products for stock planning; combine sales volume with ingredient cost and margin data before making profitability decisions.
3. Track payment mix over time to understand whether channel usage is changing.
4. Add customer IDs, product costs, discounts and store/shift identifiers to enable customer segmentation, margin analysis and more precise operational recommendations.
5. Investigate flagged transaction values against menu prices or source-system records before any correction.

Limitations

The source does not specify currency, costs, margins, customer identifiers, store location or promotional context. The analysis is descriptive and cannot establish causation. Monthly comparisons may be affected by incomplete months or the dataset's limited coverage. The report does not infer missing business facts.

Deliverables

The accompanying Jupyter Notebook contains the reproducible inspection, cleaning, EDA and visualization workflow. The cleaned CSV preserves all valid transaction rows and adds a parsed transaction timestamp.